

Jason Jaeseung Kim

Biomedical Engineering Undergraduate | Deep Learning for Medical Applications

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RESEARCH FOCUS

Applying deep learning to medical and healthcare problems, with experience in 3D medical imaging, quantitative analysis, computational MRI, and patient-facing communication.

EDUCATION

Yonsei University - B.S. Candidate in Biomedical Engineering

Mar 2021 - Expected Feb 2027

Mirae Campus | Second Major in Electrical and Electronic Engineering, Sinchon Campus

GPA	4.50 Scale	Percentile	Department Rank
4.03 / 4.30	4.18 / 4.50	97.90 / 100	1 / 49

Selected coursework: Image Processing and Applications; Digital Signal Processing; Signals and Systems; Deep Learning Laboratory; Public Health and Medical Big Data Analysis.

RESEARCH EXPERIENCE

Former Undergraduate Researcher, Advanced Image Processing Techniques Lab, Yonsei University

Dec 2023 - Feb 2026

Advisor: Prof. Joon Yul Choi | Mirae Campus

Deep Learning-Based Quantitative T1 Estimation | First-Author Manuscript in Preparation

- Designed comparative experiments across input conditions, preprocessing and harmonization strategies, and model configurations; built automated 3D preprocessing, training, baseline-reproduction, and tissue-wise evaluation pipelines.
- Contributed to a multi-year government-funded R&D proposal and literature review; prepared the conference abstract and delivered the oral presentation; drafting the first-author manuscript, figures, and tables.

Quantitative MRI Study of Overwork | First-Author Manuscript in Preparation

- Designed and executed an SPM12 voxel-wise analysis, including cohort curation, covariate adjustment, explicit masking, FWE correction, atlas-based reporting, and automated QC and group summaries.
- Prepared the conference abstract, oral presentation, figures, and tables; drafting the first-author manuscript and coordinating revisions with the supervising professor.

Extending NAMER for Learning-Enhanced MRI Motion Estimation

Ongoing Graduation Research, 2026

Graduation Research

- NAMER estimates motion from a CNN-generated image prior before model-based reconstruction; current work evaluates multiple fastMRI subjects and revises CNN training so the prior better supports downstream motion estimation.

Undergraduate Researcher, Bio-Metamaterials Lab, Yonsei University

Jul 2022 - Aug 2022

Advisor: Prof. Seung Ho Choi | Mirae Campus

- Participated in the initial installation and setup of an ellipsometer and analyzed the compositional and optical properties of biomaterials.
- Built a prototype tensile-strength measurement device and implemented a MATLAB-based user interface for measurement verification.

SELECTED RESEARCH OUTPUTS

- Oral Presentation and Outstanding Paper Award for the quantitative T1 estimation study, Korean Society of Medical and Biological Engineering Fall Conference, Nov 2025.
- Poster Presentation, NEURO2026, Jul 2026.

PROJECTS

Teacher-Difficulty-Calibrated Early-Exit Network

Undergraduate Course Project, 2026

- Undergraduate Deep Learning Laboratory course project: developed an image-classification network using knowledge distillation, teacher-difficulty guidance, dynamic early exits, and branch-fused deployment for efficient inference.

Project Lead, CareProofRecord - Deep-Learning-Based Medical Explanation Video Platform

Jun 2025 - Apr 2026

- Led the service concept and system architecture for patient-facing explanation videos designed to improve understanding and reduce information-transfer errors.
- Designed a long-context generation workflow to mitigate hallucination risk, secured KRW 1,000,000 in university startup support, and explored clinical use with regional medical institutions.

TECHNICAL SKILLS

Programming	Python; MATLAB
Medical Imaging Software	SynthStrip (skull stripping); FSL FAST (tissue segmentation); ANTs (registration and resampling); SPM12 (voxel-wise GLM and FWE correction); FreeSurfer (ROI extraction)

MERIT-BASED SCHOLARSHIPS

- National Science and Engineering Scholarship, Korea Student Aid Foundation - full tuition (Spring 2023, Fall 2023, Spring 2026).
- Hankyungbum Scholarship, Cheongpa-Hankyungbum Scholarship Foundation, Yonsei University - full tuition (Spring 2022, Fall 2022).
- Academic Excellence Scholarship, Yonsei University (Fall 2021).

HONORS AND AWARDS

Academic: Academic Merit Awards, Yonsei University, 2021-2023

Programming: Grand Prize, ASC Programming Contest, 2021

Programming: Excellence Award, Wise Coding Life Algorithm Competition, Yonsei University, 2022

Entrepreneurship: Grand Prize, MEDICI+ Startup Competition, Yonsei University Wonju LINC+ Project Group, 2021

Entrepreneurship: Honorable Mention, WAVE-LAB Startup Competition, 2025; Encouragement Award, Post Corona-19 Startup Challenge, 2022